RACE
Regular Analysis Through Continuous Exercise

Functions : RACE-01

Single Correct Question

1. The range of the function $f(x) = e^{-x} + e^{x}$, is -

(A) $f(x) \geq 1$ (B) $f(x) \leq 1$ (C) $f(x) \geq 2$ (D) $f(x) \leq 2$

MFN148

2. The range of the function $f: \mathbb{N} \to \mathbb{Z};\ f(x) = (-1)^{x-1}$, is -

(A) $[-1, 1]$ (B) $\{-1, 1\}$ (C) $\{0, 1\}$ (D) $\{0, 1, -1\}$

MFN149

Multiple Correct Question

3. Which of the following rule is/are function(s) ?

(A) $f: \left(0, \frac{\pi}{2}\right) \to [0,1],\ f(x) = \sin x$

(B) $f: \{-3, -2, -1, 0, 1, 2, 3\} \to \{1, 4, 9\}, f(x) = x^2$

(C) $f: \left[0, \frac{\pi}{2}\right) \to [0,1) \to f(x) = \cos x$

(D) $f: \left(0, \frac{\pi}{2}\right) \to (0,1) \to f(x) = \cos x$

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MFN150

4. Which of the following is (are) function ?

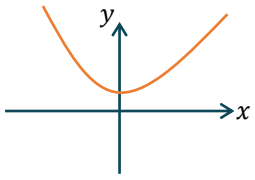
$A: \{1, 2, 3, 4\}$ $B = \{1, 4, 9, 16, 25\}$

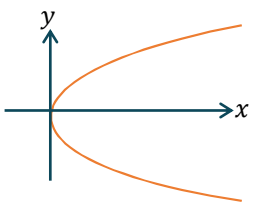
(A) $R_1 = \{(1,1)\ (2,4)\ (3,9)\ (4,16)\ (4,25)\}$ (B) $R_2 = \{(2,4)\ (3,9)\ (4,16)\}$

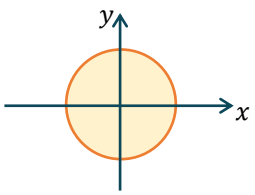
(C) $R_3 = \{(1,1)\ (2,4)\ (3,9)\ (4,16)\}$ (D) $R_4 = \{(1,1)\ (2,4)\ (3,9)\ (4,4)\}$

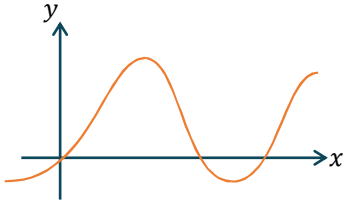
MFN151

5. Which of the following represent graph(s) of a function ?

(A) 

(B) 

(C) 

(D) 

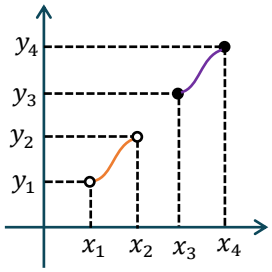
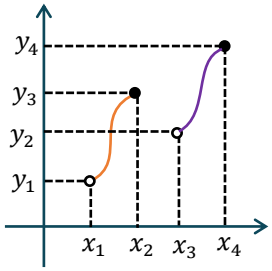
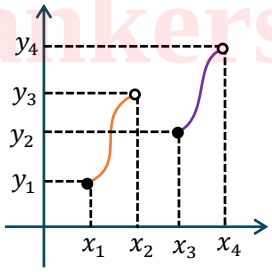
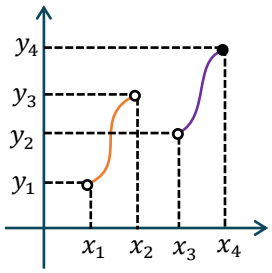
MFN152

For More Join: @IITJEE_Advanced

Comprehension Type Question

Paragraph of Question No. 6 & 7

Graph of $y = f(x)$ is drawn here, match Domain & Range of the $f(x)$

$y = f(x)$ Column-I		(Domain) Column-II		(Range) Column-III	
(P)		(I)	$(x_1, x_2] \cup (x_3, x_4]$	(i)	$(y_1, y_2) \cup [y_3, y_4]$
(Q)		(II)	$(x_1, x_2) \cup [x_3, x_4]$	(ii)	$(y_1, y_4]$
(R)		(III)	$[x_1, x_2) \cup [x_3, x_4)$	(iii)	$[y_1, y_4)$
(S)		(IV)	$(x_1, x_2) \cup (x_3, x_4]$	(iv)	$(y_1, y_4) - \{y_2\}$

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6. Select the correct match

(A) P-(II)-(ii) (B) Q-(I)-(ii) (C) R-(III)-(ii) (D) S-(II)-(iv)

MFN153

7. Select the correct match

(A) P-(II)-(i) (B) Q-(II)-(i) (C) R-(III)-(ii) (D) S-(I)-(ii)

MFN154

For More Join: @IITJEE_Advanced

Matrix Match Type Question

8. Match the function given in **column-I** with corresponding domain in **column-II**.

	Column-I		**Column-II**
(P)	$y=\sqrt{\frac{x(x-1)^3(x-5)}{(x+4)^2(x-2)}}$	(1)	$x \in [-2, 0] \cup [2, \infty)$
(Q)	$y=\sqrt{(e^x-1)(x^2-4)}$	(2)	$-1 \le x \le 1$
(R)	$y=\sqrt{-px}\ \ (p>0)$	(3)	$x \in (-\infty, -4) \cup (-4, 0] \cup [1, 2) \cup [5, \infty)$
(S)	$y=\sqrt{1-\lvert x\rvert}$	(4)	$-\infty < x \le 0$

MFN155

9. Match the function given in **column-I** with corresponding domain in **column-II** .

	Column-I		**Column-II**
(P)	$f(x)=\sqrt{\lvert x\rvert - x}$	(1)	$x \in \left(-\infty, -\frac{1}{2}\right] \cup (0,1) \cup (2,\infty)$
(Q)	$f(x)=\frac{1}{\sqrt{\lvert x\rvert - x}}$	(2)	$x \in (-\infty, 0) \cup (4, \infty)$
(R)	$f(x)=\sqrt{\frac{(2x+1)}{x^3-3x^2+2x}}$	(3)	$x \in (-\infty, 0)$
(S)	$f(x)=\frac{1}{\sqrt{x^2-4x}}$	(4)	$x \in \mathbb{R}$

MFN156

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10. Match the function given in **column-I** with corresponding domain in **column-II** .

	Column-I		**Column-II**
(P)	$y=\frac{3}{4-x^2}+\log_{10}(x^3-x)$	(1)	$2k\pi < x < (2k+1)\pi$, where k is an integer
(Q)	$y=\frac{1}{\sqrt{\sin x}}+\sqrt[3]{\sin x}$	(2)	$4 \le x \le 6$
(R)	$y=\log_{10}\left(\sqrt{x-4}+\sqrt{6-x}\right)$	(3)	$2 < x < 3$
(S)	$y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$	(4)	$-1 < x < 0$ or $1 < x < 2$ or $2 < x < \infty$

MFN157

11. Match the function given in **column-I** with corresponding domain in **column-II** .

	Column-I		**Column-II**
(P)	$y = \log_x 2$	(1)	$-2 \le x < 0$ or $0 < x < 1$
(Q)	$y=\frac{1}{\log_{10}(1-x)}+\sqrt{x+2}$	(2)	$\frac{3}{2} < x < 2$ or $2 < x < \infty$
(R)	$y=\sqrt{x}+\sqrt[3]{\frac{1}{x-2}}-\log_{10}(2x-3)$	(3)	$x \in \mathbb{R}$
(S)	$y=\frac{1}{x^2+1}$	(4)	$0 < x < 1$ or $1 < x < \infty$

MFN158

For More Join: @IITJEE_Advanced

Integer Type Question

12. If $f(x) = 3x + 4$; $g(x) = \begin{cases} x^2+1; & x \geq 1 \\ 2x+3; & x < 1 \end{cases}$ and $h(x) = \begin{cases} e^x & ; \quad x \geq 5 \\ 2x^2+3x & ; \quad 2 < x < 5 \\ 5-x & ; \quad x < 2 \end{cases}$ then find $h(f(g(-2)))$

MFN159

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	C	B	A,D	C,D	A,D	B	A	P→3;Q→1; R→4;S→2
Question	9			10			11	12
Answer	P→4;Q→3;R→1;S→2			P→4;Q→1;R→2;S→3			P→4;Q→1;R→2;S→3	4

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Functions : RACE-02

Single Correct Question

1. Arrange the following in decreasing order $3^x, 5^x, 2^x, 4^x$ if $x > 0$

(A) $3^x > 5^x > 2^x > 4^x$ (B) $5^x > 4^x > 3^x > 2^x$

(C) $2^x > 3^x > 4^x > 5^x$ (D) $2^x > 5^x > 4^x > 3^x$

MFN160

2. Find the domain of the function $f(x)=\frac{1}{\sqrt{[|x|-7]-11}}$, where [.] denotes greatest integer function.

(A) $x \in (-\infty, -17] \cup [19,\infty)$ (B) $x \in (-\infty, -19] \cup [17,\infty)$

(C) $x \in (-\infty, -19] \cup [19,\infty)$ (D) $x \in (-\infty, -20] \cup [19,\infty)$

MFN161

3. If $y = 2[x] + 1 = 3[x + 1] - 5$ then find $[x + 2y]$, where [.] denotes greatest integer function.

(A) 17 (B) 18 (C) 19 (D) 20

MFN162

4. Find the range of x if $[x]^2 - [x] - 2 = 0$, where [.] denotes greatest integer function.

(A) $x \in [1, 0) \cup [2, 3)$ (B) $x \in [-1, 0) \cup [2, 3)$

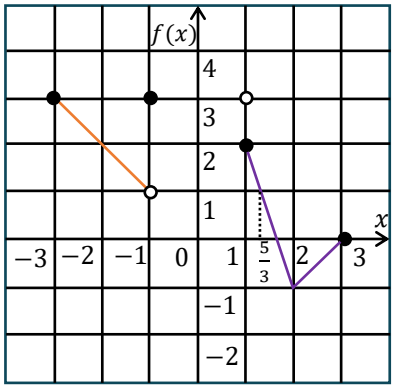
(C) $x \in [-1, 0) \cup [-2, 3)$ (D) $x \in [-1, 0] \cup [2, 3)$

MFN163

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Comprehension Type Question

Solve the following inequalities using graph of $f(x)$:



5. Find the values of x for $0 \le f(x) \le 1$

(A) $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$ (B) $[-2, -1) \cup [1, 3]$ (C) $[-3, -2] \cup [-1, 1]$ (D) $\left(\frac{5}{3}, 3\right) - \{2\}$

MFN164

6. Find the values of x for $-1 \le f(x) \le 2$

(A) $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$ (B) $[-2, -1) \cup [1, 3]$ (C) $[-3, -2] \cup [-1, 1]$ (D) $\left(\frac{5}{3}, 3\right) - \{2\}$

MFN165

For More Join: @IITJEE_Advanced

7. Find the values of x for $f(x) > -1$ & $f(x) < 0$

(A) $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$ (B) $[-2, -1) \cup [1, 3]$ (C) $[-3, -2] \cup [-1, 1]$ (D) $\left(\frac{5}{3}, 3\right) - \{2\}$

MFN166

Matrix Match Type Question

8.

	Column-I	**Column-II**
(P)	Find range of the following functions $f: R \to R, f(x) = x^2 - 8x + 7$	(1) $[-8, 16]$
(Q)	Find range of the following functions $f: [-1, 3] \to R, f(x) = x^2 - 8x + 7$	(2) $[-8, \infty)$
(R)	Find range of the following functions $f: [-1, \infty) \to R, f(x) = x^2 - 8x + 7$	(3) $[-9, \infty)$
(S)	Find range of the following functions $f: (-\infty, 3] \to R, f(x) = x^2 - 8x + 7$	(4) $[16, \infty)$

MFN167

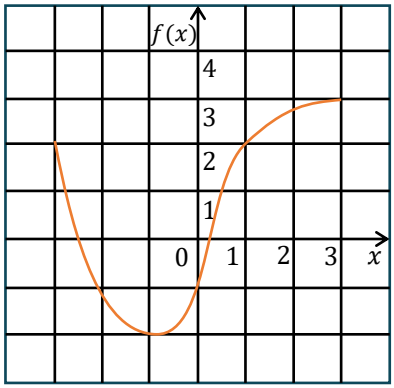
9. Find the range of the following functions :

	Column-I	**Column-II**
(P)	$f(x) = 2^{x^2} + 1$	(1) $\left[\frac{1}{11}, \frac{1}{5}\right]$
(Q)	$f(x) = \frac{1}{8 - 3\sin x}$	(2) $\left[\frac{1}{3}, 1\right]$
(R)	$f(x) = \frac{1}{2 - \cos 3x}$	(3) $(0, 1]$
(S)	$f(x) = \frac{1}{x^2 + 1}$	(4) $[2, \infty)$

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MFN168

10. The graph of a function f is given.



	Column-I	**Column-II**
(P)	State the value of $f(-1)$.	(1) $-3, 1$
(Q)	For what values of x is $f(x) = 2$	(2) $[-1, 3]$
(R)	State the domain and range of f.	(3) $[-3, 3], [-2, 3]$
(S)	On what interval is f increasing ?	(4) -2

MFN169

For More Join: @IITJEE_Advanced

Subjective Type Question

11. Draw the graph of following

(i) $y = |x+3| + |x-1| + |x-5|$

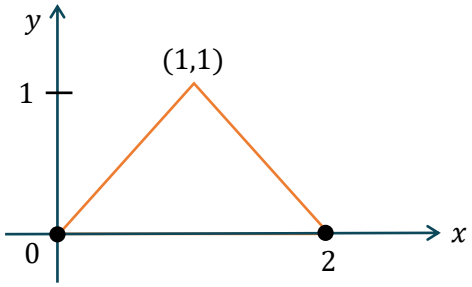
(ii) $f(x) = [e^x]$, where [.] denotes greatest integer function.

(iii) $f(x) = \begin{cases} x+1; & -1 \le x < 4 \\ -x; & 4 \le x < 7 \end{cases}$

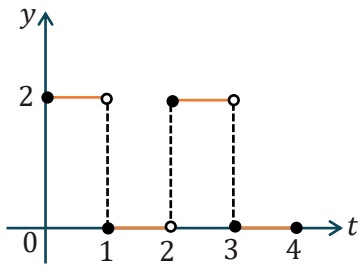
MFN170

12. Find a formula for each function graphed

(i)



(ii)



MFN171

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	B	C	A	B	A	B	D	P→3, Q→1, R→3, S→2
Question	9					10		
Answer	P→4, Q→1, R→2, S→3					P→4, Q→1, R→3, S→2		

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Functions : RACE-03

Single Correct Question

1. If $[x]$ and $\{x\}$ denotes the greatest integer function less than or equal to x and fractional part function respectively, then the number of real x, satisfying the equation $(x-2)[x]=\{x\}-1$, is-

(A) 0 (B) 1 (C) 2 (D) infinite

MFN172

2. If $[a]$ denotes the greatest integer less than or equal to a and $-1 \le x < 0, 0 \le y < 1, 1 \le z < 2$, then $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is equal to -

(A) $[x]$ (B) $[y]$ (C) $[z]$ (D) none of these

MFN173

3. The domain of definition of $f(x)=\frac{\log_2(x+3)}{x^2+3x+2}$ is :

(A) $R\backslash\{-1,-2\}$ (B) $(-2,\infty)$ (C) $R\backslash\{-1,-2,-3\}$ (D) $(-3,\infty)\backslash\{-1,-2\}$

MFN174

4. The number of integers lying in the domain of the function $f(x)=\sqrt{\log_{0.5}\left(\frac{5-2x}{x}\right)}$ is -

(A) 3 (B) 2 (C) 1 (D) 0

MFN175

Comprehension Type Question

Paragraph (Q. 5 to 7)

Let $\{x\}$ & $[x]$ denote the fractional and integral part of a real number x respectively.

5. Find x if $2x+\{x\}=5[x]$

(A) $x \in \{0\}$ (B) $x \in \{0,1\}$ (C) $x \in \{1,2\}$ (D) $x \in \{0,2\}$

MFN176

6. Find x if $x-\{x\}=2[x]$

(A) $x \in [-1,1)$ (B) $x \in [0,2)$ (C) $x \in [0,1)$ (D) $x \in [0,1]$

MFN177

7. Find the number of values of x if $4\{x\}=x+[x]$.

(A) 1 (B) 2 (C) 3 (D) 4

MFN178

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For More Join: @IITJEE_Advanced

Matrix Match Type Question

8. Let $\{x\}$ & $[x]$ denotes the fraction and integral part of a real number x respectively, then match the column.

	Column-I		**Column-II**
(P)	$[x^2] \geq 4$	(1)	$x \in [2, 4)$
(Q)	$[x]^2 - 5[x] + 6 = 0$	(2)	$x \in (-\infty, -2] \cup [2, \infty)$
(R)	$x = \{x\}$	(3)	$x \in (-\infty, -5)$
(S)	$[x] < -5$	(4)	$x \in \{-2\}$
		(5)	$x \in [0, 1)$

MFN179

9. Find the range of the following functions :

	Column-I		**Column-II**
(P)	$f(x)=e^{(x-1)^2}$	(1)	$[0,\infty)$
(Q)	$f(x) = x^3 - x^2 + x + 1$	(2)	$R-\left\{\frac{7}{3}\right\}$
(R)	$f(x) = \log(x^8 + x^4 + x^2 + 1)$	(3)	$\mathbb{R}$
(S)	$f(x)=\frac{7x+9}{3x-5}$	(4)	$[1, \infty)$

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MFN180

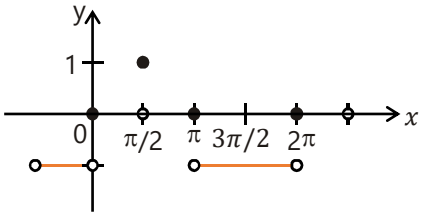
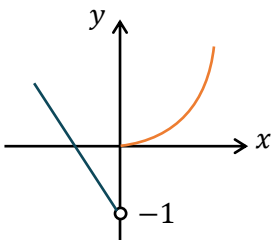
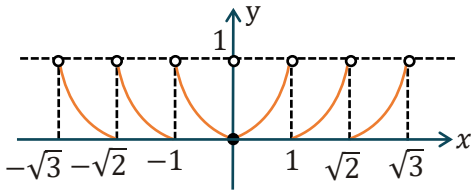
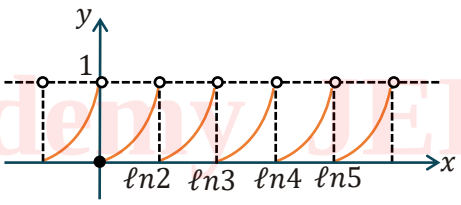
10. Find the domains of definitions of the following functions :

	Column-I		**Column-II**
(P)	$f(x)=\log_{3-2\{x\}}(x^2-5x+13)$	(1)	$x \in (-\infty,-3)\cup[2,\infty)$
(Q)	$f(x)=\sqrt{\frac{9-3^x}{5^{-x}-125}}$	(2)	$\{4\} \cup [5, \infty)$
(R)	$f(x)=\sqrt{x^2-\lvert x\rvert}+\frac{1}{\sqrt{9-x^2}}$	(3)	$x \in R$
(S)	$f(x)=\sqrt{(x^2-3x-10)ln^2(x-3)}$	(4)	$(-3, -1] \cup\{0\} \cup [1, 3)$

MFN181

For More Join: @IITJEE_Advanced

11. Match the graph of function given in **Column-I** with sketch in **Column-II :**

Column-I		**Column-II**	
(P)	$f(x)=\begin{cases} x^2; & x\geq 0 \\ -x-1; & x<0 \end{cases}$	(1)	
(Q)	$f(x)=[\sin x]$	(2)	
(R)	$f(x)=\{e^x\}$	(3)	
(S)	$f(x)=\{x^2\}$	(4)	

MFN182

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Integer Type Question

12. Find the number of integer value of x in domain of function :

$$f(x)=\sin\left(\sqrt{1-x^2}\right)+\sqrt{x+2}+\frac{1}{\log_{10}(x+1)}$$

MFN183

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Answer	D	C	D	C	A	C	B	P→1,2,3,4;Q→1,2;R→5;S→2,3

Question	**9**	**10**	**11**	**12**
Answer	P→4;Q→3;R→1;S→2	P→3;Q→1;R→4;S→2	P→2;Q→1;R→4;S→3	1

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Functions : RACE-04

Single Correct Question

1. A function f has domain $[-1, 2]$ and range $[0, 1]$. The domain and range respectively of the function g defined by $g(x) = 1 - f(x + 1)$ is

(A) $[-1, 1]$; $[-1, 0]$ (B) $[-2, 1]$; $[0, 1]$ (C) $[0, 2]$; $[-1, 0]$ (D) $[1, 3]$; $[-1, 0]$

MFN184

2. The range of the function $f(x)=\text{sgn}\left(\frac{\sin^2 x+2\sin x+4}{\sin^2 x+2\sin x+3}\right)$ is (where sgn(.) denotes signum function)-

(A) $\{-1,0,1\}$ (B) $\{-1,0\}$ (C) $\{1\}$ (D) $\{0,1\}$

MFN185

3. The range of the function $f(x)=\sqrt{4-x^2}+\sqrt{x^2-1}$ is

(A) $\left[\sqrt{3},\sqrt{7}\right]$ (B) $\left[\sqrt{3},\sqrt{5}\right]$ (C) $\left[\sqrt{2},\sqrt{3}\right]$ (D) $\left[\sqrt{3},\sqrt{6}\right]$

MFN186

4. Range of the function $f(x)=\frac{x^2+x+2}{x^2+x+1}$ is -

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(A) $[1, 2]$ (B) $[1, \infty)$ (C) $\left[2,\frac{7}{3}\right]$ (D) $\left(1,\frac{7}{3}\right]$

MFN187

5. Range of the function $f(x)=\frac{\{x\}}{1+\{x\}}$, where $\{x\}$ denotes the fractional part function, is

(A) $[0, 1)$ (B) $\left[0,\frac{1}{2}\right]$ (C) $\left[0,\frac{1}{2}\right)$ (D) $\left(0,\frac{1}{2}\right)$

MFN188

6. Which of the following function is implicit function ?

(A) $xy - \cos(x + y) = 0$ (B) $y = x^3$

(C) $y = \log(x^2 + x + 1)$ (D) $y = |x|$

MFN189

7. For the function $f(x)=\frac{e^x+1}{e^x-1}$, if $n(d)$ denotes the number of integers which are not in its domain and $n(r)$ denotes the number of integers which are not in its range, then $n(d) + n(r)$ is equal to -

(A) 2 (B) 3 (C) 4 (D) Infinite

MFN190

8. Find range of $f(x)=\frac{2^x-2^{-x}}{2^x+2^{-x}}$

(A) $(-1, 1)$ (B) $[-1, 1]$ (C) $\{-1, 1\}$ (D) $(-1, 1]$

MFN191

For More Join: @IITJEE_Advanced

9. Find range of $f(x)=\dfrac{(\ell n\, x)^2-5\ell n\, x+4}{(\ell n\, x)^2-2\ell n\, x-3}$

(A) $\mathbb{R}^+$ (B) $\mathbb{R}^-$ (C) $\mathbb{R}$ (D) $\mathbb{R}_0$

MFN192

Multiple Correct Question

10. The range of the function $f(\theta)=\sqrt{8\sin^2\theta+4\cos^2\theta-8\sin\theta\cos\theta}$ is -

(A) $\left[\sqrt{5}-1,\sqrt{5}+1\right]$ (B) $\left[0,\sqrt{5}+1\right]$

(C) $\left[\sqrt{6-\sqrt{20}},\sqrt{6+\sqrt{20}}\right]$ (D) none of these

MFN193

Matrix Match Type Question

11. Find the range of the following function :

Column-I		**Column-II**	
(P)	$\log_4\left\|x+\dfrac{1}{x}\right\|$	(1)	$(-\infty-1]\cup(0,\infty)-\left\{\dfrac{-3}{2}\right\}$
(Q)	$f(x)=\sin(3x^2+1)$	(2)	$R-\left\{\dfrac{1}{4},\dfrac{-1}{11}\right\}$
(R)	$f(x)=\dfrac{x^2-5x+6}{4x^2-5x-6}$	(3)	$\left[\dfrac{1}{2},\infty\right)$
(S)	$f(x)=\dfrac{3x-6}{x^3-4x^2+2x+4}$	(4)	$[-1,1]$

MFN194

Integer Type Question

12. Let $f(x)=(x+1)(x+2)(x+3)(x+4)+5$ where $x\in[-6,6]$. If the range of the function is $[a,b]$ where $a,b\in N$ then find the value of $(a+b)$.

MFN195

Pankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	D	D	C	A	C	A	C	A,C
Question	11				12					
Answer	P→3;Q→4;R→2;S→1				5049					

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Functions : RACE-05

Single Correct Question

1. If $2f(x) - 3f\left(\frac{1}{x}\right) = x^2, x$ is not equal to zero, then $f(2)$ is equal to-

(A) $-\frac{7}{4}$ (B) $\frac{5}{2}$ (C) –1 (D) none of these

MFN196

2. If $f(x) = \frac{4^x}{4^x + 2}$, then $f(x) + f(1 - x)$ is equal to-

(A) 0 (B) –1 (C) 1 (D) 4

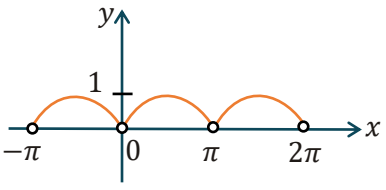
MFN197

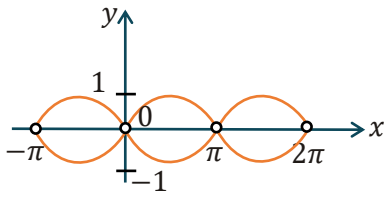
3. If $x^4 \ f(x) - \sqrt{1 - \sin 2\pi x} = |f(x)| - 2f(x)$, then $f(-2)$ equals

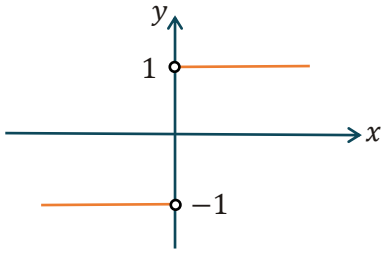
(A) $\frac{1}{17}$ (B) $\frac{1}{11}$ (C) $\frac{1}{19}$ (D) 0

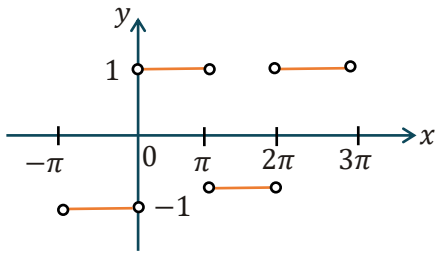
MFN198

4. Which of the following is the graph of $y = \frac{|\sin x|}{\sin x}$?

(A)


(B)


(C)


(D)


MFN199

5. A lion moves in the region given by the graph $y - |y| - x + |x| = 0$. Then on which of the following curve a person can move so that he does not encounter lion -

(A) $y = e^{-|x|}$ (B) $y = \frac{1}{x}$ (C) $y = \text{signum}(x)$ (D) $y = -|4 + |x||$

MFN200

For More Join: @IITJEE_Advanced

6. Let function $f: R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is -

(A) one to one and onto

(B) one to one but not onto

(C) onto but not one to one

(D) neither one to one nor onto

MFN201

7. Let $f(x)=\frac{x}{1+x}$ defined from $(0,\infty) \to [0,\infty)$ then by $f(x)$ is -

(A) one-one but not onto

(B) one-one and onto

(C) many one but not onto

(D) many one and onto

MFN202

8. Let $f: N \to N$ be defined as $f(n)=\begin{cases}\frac{n+1}{2}, & \text{if } n \text{ is odd}\\ \frac{n}{2}, & \text{if } n \text{ is even}\end{cases}$ for all $n \in N$. Then f is-

(A) injective but not surjective

(B) surjective but not injective

(C) both injective as well as surjective

(D) neither injective nor surjective

MFN203

Rankers Academy JEE

9. Let $E = \{1,2,3,4\}$ & $F = \{1,2\}$. Then the number of onto functions from E to F is -

(A) 14 (B) 16 (C) 12 (D) 8

MFN204

Multiple Correct Question

10. If a function is defined by an implicit equation $2^{|x|+|y|} + 2^{|x|-|y|} = 2$, then -

(A) Domain of function is singleton

(B) Range of function is singleton

(C) graph of the function intersects the line $y = x$

(D) maximum value of function is 2

MFN205

11. Which of the following pair of functions are identical?

(A) $f(x) = sgn(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$

(B) $f(x) = \sqrt{\frac{1-\cos 2x}{1+\cos 2x}}$ and $g(x) = \tan x$

(C) $f(x) = \ell n(1+x) + \ell n(1-x)$ and $g(x) = \ell n(1-x^2)$

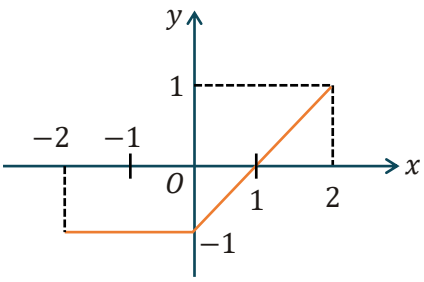
(D) $f(x)=\frac{\cos x}{1-\sin x}$ and $g(x)=\frac{1+\sin x}{\cos x}$

MFN206

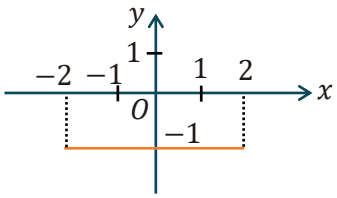
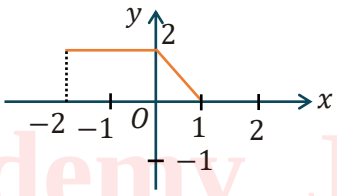
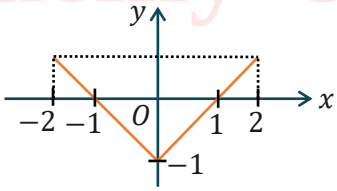
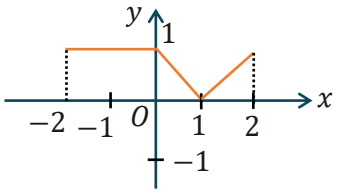
For More Join: @IITJEE_Advanced

Matrix Match Type Question

12. The graph of the function $y = f(x)$ is as follows :



Match the function mentioned in **Column-I** with the respective graph given in **Column-II**.

Column-I		**Column-II**	
(P)	$y = \|f(x)\|$	(1)	
(Q)	$y = f(\|x\|)$	(2)	
(R)	$y = f(-\|x\|)$	(3)	
(S)	$y = (\|f(x)\| - f(x))$	(4)	

Rankers Academy JEE

MFN207

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	A	D	D	A	A	B	A	A,B,C
Question	11	12								
Answer	A,C	P→4; Q→3;R→1; S→2								

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Functions : RACE-06

Single Correct Question

1. Which of the following function is surjective but not injective

(A) $f:\mathbb{R}\to\mathbb{R}\ f(x)=x^4+2x^3-x^2+1$ (B) $f:\mathbb{R}\to\mathbb{R}f(x)=x^3+x+1$

(C) $f:\mathbb{R}\to\mathbb{R}^+\ f(x)=\sqrt{1+x^2}$ (D) $f:\mathbb{R}\to\mathbb{R}\ f(x)=x^3+2x^2-x+1$

MFN208

2. A mapping $f:A\to[-1,1]$ defined by $f(x)=\sin x,\ \forall\ x\in\mathbb{R}$, where A is a subset of $\mathbb{R}$ (the set of all real numbers) is one-one and onto if A is the interval, then A belongs to

(A) $[0,2\pi]$ (B) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (C) $[-\pi,\pi]$ (D) $[0,\pi]$

MFN209

3. Let $f(x)=\frac{\alpha x}{x+1}$, $\forall x\in D_f$,Then for what value of α is $f(f(x))=x$?

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) –1

MFN210

Rankers Academy JEE

4. Let $g(x)=1+x-[x]$ & $f(x)=\begin{cases}-1\ , & x<0\\ 0\ , & x=0\\ 1\ , & x>0\end{cases}$. Then for all x, $f(g(x))$ is equal to,

where [.] denotes greatest integer function.

(A) x (B) 1 (C) $f(x)$ (D) $g(x)$

MFN211

5. $f(x)=x^3-x$ & $g(x)=\sin 2x$, then select the incorrect option-

(A) $f(f(1))=0$ (B) $f(f(-1))=0$ (C) $f\left(g\left(\frac{\pi}{2}\right)\right)=0$ (D) $f\left(g\left(\frac{\pi}{4}\right)\right)=1$

MFN212

6. If f be the greatest integer function and g be the modulus function, then

$(gof)\left(-\frac{5}{3}\right)-(fog)\left(-\frac{5}{3}\right)=$

(A) 1 (B) – 1 (C) 2 (D) 4

MFN213

7. A function $f:\mathbb{R}\to\mathbb{R}$ is such that $f\left(\frac{1-x}{1+x}\right)=x$ for all $x\neq-1$. Then which of the following not correct.

(A) $f(f(x))=x$ (B) $f\left(\frac{1}{x}\right)=-f(x),x\neq0$

(C) $f(-x-2)=-f(x)-2.$ (D) $f(-x-2)=f(x)-2.$

MFN214

For More Join: @IITJEE_Advanced

Multiple Correct Question

8. $f(x)=\begin{cases}[x], & -2\le x\le -1\\ |x|+1, & -1<x\le 2\end{cases}$ and $g(x)=\begin{cases}[x], & -\pi\le x\le 0\\ \sin x, & 0\le x\le \pi\end{cases}$, [.] denotes the greatest integer function. If $h(x)=g(f(x))$

(A) domain of $h(x)$ is $[-1, 1]$

(B) Range of $h(x)$ is $\{-2, -1\}\cup[\sin 3, 1]$

(C) Domain of $h(x)$ is $[-2, 2]$

(D) Range of $h(x)$ is $\{-2, -1\}\cup[\sin 2, 1]$

MFN215

Matrix Match Type Question

9. Match the functions given in column-I correctly with mappings given in column-II.

Column-I	**Column-II**
(P) $f:\left[-\frac{1}{2},\frac{1}{2}\right]\to\left[\frac{4}{7},\frac{4}{3}\right]$, $f(x)=\frac{1}{x^2+x+1}$	(1) Injective mapping
(Q) $f:[-2,2]\to[-1,1]$, $f(x)=\sin x$	(2) Non-injective mapping
(R) $f:\mathbb{R}-I\to\mathbb{R}$ $f(x)=\ell n\{x\}$, (where {.} represents fractional part function)	(3) Surjective mapping
(S) $f:(-\infty,0]\to[1,\infty)$, $f(x)=(1+\sqrt{-x})+(\sqrt{-x}-x)$	(4) Non-surjective mapping
	(5) Bijective mapping

Rankers Academy JEE

MFN216

Integer Type Question

10. If $f(x)=\begin{cases}0 & x<1\\ 2x-2 & x\ge 1\end{cases}$; then the number of solutions of the equation $f(f(f(x)))=x$ is

MFN217

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	D	B	D	A	D	B,C	P→1,3,5; Q→2,3; R→2,4; S→1,3,5	2

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Functions : RACE-07

Single Correct Question

1. If $f:(-\infty,3]\to[7,\infty)$; $f(x)=x^2-6x+16$, then which of the following is true -

(A) $f^{-1}(x)=3+\sqrt{x-7}$ (B) $f^{-1}(x)=3-\sqrt{x-7}$

(C) $f^{-1}(x)=\dfrac{1}{x^2-6x+16}$ (D) f is many-one

MFN218

2. If $f:[1,\infty)\to[2,\infty)$ is given by, $f(x)=x+\dfrac{1}{x}$, then $f^{-1}(x)$ equals -

(A) $\dfrac{x+\sqrt{x^2-4}}{2}$ (B) $\dfrac{x}{1+x^2}$ (C) $\dfrac{x-\sqrt{x^2-4}}{2}$ (D) $1-\sqrt{x^2-4}$

MFN219

3. Suppose $f(x)=(x+1)^2$ for $x\geq -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y=x$, then $g(x)$ equals -

(A) $-\sqrt{x}-1, x\geq 0$ (B) $\dfrac{1}{(1+x)^2}$, $x\geq -1$ (C) $\sqrt{x+1}$, $x\geq -1$ (D) $\sqrt{x}-1, x\geq 0$

Rankers Academy JEE

MFN220

4. $f:\mathbb{R}\to\mathbb{R}$ such that $f(x)=\ell n\left(x+\sqrt{x^2+1}\right)$. Another function $g(x)$ is defined such that $gof(x)=x,\ \forall x\in R$. Then $g(2)$ is -

(A) $\dfrac{e^2+e^{-2}}{2}$ (B) e^2 (C) $\dfrac{e^2-e^{-2}}{2}$ (D) e^{-2}

MFN221

5. If $f(x)=x|x|$ then $f^{-1}(x)$ equals-

(A) $\sqrt{|x|}$ (B) $(sgn\ x).\sqrt{|x|}$ (C) $-\sqrt{|x|}$ (D) Does not exist

(where sgn(x) denotes signum function of x)

MFN222

6. Let f be a function defined by $f(x)=(x-1)^2+1, (x\geq 1)$

Statement - 1 : The set $\{x: f(x)=f^{-1}(x)\}=\{1,2\}$

Statement - 2 : f is bijection and $f^{-1}(x)=1+\sqrt{x-1}$, $x\geq 1$.

(A) Statement–1 is true, Statement–2 is false.

(B) Statement–1 is false, Statement–2 is true.

(C) Statement–1 is true, Statement–2 is true ; Statement–2 is a correct explanation for Statement–1.

(D) Statement–1 is true, Statement–2 is true ; Statement–2 is not a correct explanation for statement–1.

MFN223

For More Join: @IITJEE_Advanced

7. If $f(x) = 3x - 5$, then $f^{-1}(x)$ -

(A) is given by $\frac{1}{3x-5}$ (B) is given by $\frac{x+5}{3}$

(C) does not exist because f is not one–one (D) does not exist because f is not onto

MFN224

8. If the function $f: [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is -

(A) $\left(\frac{1}{2}\right)^{x(x-1)}$ (B) $\frac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$

(C) $\frac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (D) Not defined

MFN225

Matrix Match Type Question

9.

	Column-I		**Column-II**
(P)	$f(x) = \frac{x(a^x+1)}{a^x-1}$	(1)	Neither even nor odd function
(Q)	$f(x) = x\sin^2 x - x^3$	(2)	Even function
(R)	$f(x) = \frac{x}{e^x-1} + \frac{x}{2} + 1$	(3)	Odd function
(S)	$f(x) = [(x+1)^2]^{\frac{1}{3}} + [(x-1)^2]^{\frac{1}{3}}$	(4)	Identity function

MFN226

Rankers Academy JEE

Integer Correct Type Question

10. Let a and b be real numbers and let $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$.
If $f(\log_{10}(\log_3 10)) = 5$ then find the value of $f(\log_{10}(\log_{10} 3))$.

MFN227

11. Let $f(x)$ be a periodic function with period $'p'$ satisfying
$f(x) + f(x+3) + f(x+6) + \ldots + f(x+42) =$ constant $\forall\, x \in \mathbb{R}$, then sum of digits of $'p'$ is

MFN228

12. Find the inverse of $f(x) = 2^{\log_{10} x} + 8$ and hence solve the equation $f(x) = f^{-1}(x)$.

MFN229

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	B	A	D	C	B	C	B	B	P→2;Q→3; R→2;S→2	3
Question	**11**	**12**								
Answer	9	10								

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Functions : RACE-08

Single Correct Question

1. The fundamental periods (if periodic) of the function, $f(x) = \ell n(\cos x) + \tan^3 x$ is

(A) π (B) 2π (C) 3π (D) 4π

MFN230

2. The fundamental periods (if periodic) of the function, $f(x) = e^{x-[x]}$, [.] denotes greatest integer function is

(A) 1 (B) 2 (C) 3 (D) 4

MFN231

3. The fundamental periods (if periodic) of the function, $f(x) = \left|\sin\frac{x}{2}\right| + \left|\cos\frac{x}{2}\right|$

(A) π (B) 2π (C) 3π (D) 4π

MFN232

Matrix Match Type Question

4. Match the function mentioned in column-I with the respective classification given in column-II. (where [.] and {.} denote greatest integer function and fractional part function respectively)

Column-I		Column-II
(P)	$f: R \to R^+$ $f(x) = (e^{[x]})(e^{\{x\}})$	(1) one-one
(Q)	$f: (-\infty,-2) \cup (0,\infty) \to R$ $f(x) = \ell n(x^2 + 2x)$	(2) many-one
(R)	$f: [-2,2] \to [-1,1]$ $f(x) = \sin x$	(3) onto
(S)	$f: R \to R$ $f(x) = x^3 - 3x^2 + 3x - 7$	(4) periodic
		(5) aperiodic

MFN233

Integer Type Question

5. Suppose that f is an even, periodic function with period 2, and that $f(x) = x$ for all x in interval [0,1]. Find the value of $f(3.14)$.

MFN234

ANSWER KEY

Question	1	2	3	4	5
Answer	B	A	A	P→1,3,5; Q→2,3,5; R→2,3,5; S→1,3,5	0.86

For More Join: @IITJEE_Advanced